

Experiment 03:

7. Laboratory Exercise

1. Verify the network connectivity of the Kali Linux virtual machine by using an appropriate command and record the output.

```
(student@kali)-[~/Shivam]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a4:44:ea brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86293sec preferred_lft 86293sec
    inet6 fd17:625c:f037:2:f5e6:38da:45aa:2568/64 scope global temporary dynamic
        valid_lft 86295sec preferred_lft 14295sec
    inet6 fd17:625c:f037:2:a00:27ff:fea4:44ea/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86295sec preferred_lft 14295sec
    inet6 fe80::a00:27ff:fea4:44ea/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
(student@kali)-[~/Shivam]
$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.017 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.048 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.054 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.046 ms
^C
— 127.0.0.1 ping statistics —
7 packets transmitted, 7 received, 0% packet loss, time 6142ms
rtt min/avg/max/mdev = 0.017/0.043/0.054/0.011 ms
```

Observation:

The **ip a** command displayed the network configuration of the Kali Linux virtual machine, including the loopback (**lo**) and Ethernet (**eth0**) interfaces. The **eth0** interface was assigned a valid IP address, confirming that the virtual machine was connected to the network. The **ping 127.0.0.1** command returned replies with **0% packet loss**, showing that the local network stack was functioning correctly and the system had proper network connectivity.

2. Perform a basic scan of the target system using Nmap and identify the open ports and associated services.

```
(student@kali)-[~/Shivam]
$ nmap github.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-31 11:27 +0530
Nmap scan report for github.com (20.207.73.82)
Host is up (0.0013s latency).
Other addresses for github.com (not scanned): 64:ff9b::14cf:4952
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
110/tcp   open  pop3
143/tcp   open  imap
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 9.37 seconds
```

Observation:

The basic Nmap scan successfully detected the target host and identified the open ports running on it. The scan showed that ports **22 (SSH)**, **80 (HTTP)**, **110 (POP3)**, **143 (IMAP)**, and **443 (HTTPS)** were open, while the remaining ports were filtered. This indicates that the target system was reachable and providing multiple network services.

3. Perform service version detection on the target system and identify the versions of the running services.

```
(student@kali)-[~/Shivam]
$ sudo nmap -sV github.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-31 11:32 +0530
Nmap scan report for github.com (20.207.73.82)
Host is up (0.0031s latency).
Other addresses for github.com (not scanned): 64:ff9b::14cf:4952
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
22/tcp    open  ssh      (protocol 2.0)
80/tcp    open  http
110/tcp   open  pop3?
143/tcp   open  imap?
443/tcp   open  ssl/https

3 services unrecognized despite returning data. If you know the service/version, please submit th
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 167.82 seconds
```

Observation:

Using the **-sV** option, Nmap attempted to determine the versions of the services running on the detected open ports. The scan successfully identified the SSH service as **protocol 2.0**, while other services such as HTTP, POP3, IMAP, and HTTPS were detected but some version

details could not be fully identified. This shows that service version detection provides additional information beyond a normal port scan.

4. Scan the first 100 ports of the target system and list the open ports identified.

```
(student@kali)-[~/Shivam]
$ sudo nmap -p 1-100 github.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-31 11:49 +0530
Nmap scan report for github.com (20.207.73.82)
Host is up (0.0013s latency).
Other addresses for github.com (not scanned): 64:ff9b::14cf:4952
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 6.44 seconds
```

Observation:

The scan of ports 1–100 found only **port 22 (SSH)** and **port 80 (HTTP)** to be open. All other ports within this range were filtered or closed. This confirms that only a few commonly used services were accessible among the first hundred ports.

5. Perform an aggressive scan on the target system to gather detailed information about the operating system, services, and network characteristics.

```
(student@kali)-[~/Shivam]
$ sudo nmap -A github.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-31 11:39 +0530
Nmap scan report for github.com (20.207.73.82)
Host is up (0.0020s latency).
Other addresses for github.com (not scanned): 64:ff9b::14cf:4952
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          (protocol 2.0)
|_ fingerprint-strings:
|   NULL:
|_   SSH-2.0-10bb30f
80/tcp    open  http
|_ fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.1 301 Moved Permanently
|     Content-Length: 0
|     Location: https://nice%20ports%2C/Tri%6Eity.txt%2ebak
|     Via: HTTP/1.1 forward.http.proxy:3128
|     Connection: close
|_   GetRequest, HTTPOptions:
|     HTTP/1.1 301 Moved Permanently
|     Content-Length: 0
|     Location: https://
|     Via: HTTP/1.1 forward.http.proxy:3128
|     Connection: close
|_   TerminalServer, TerminalServerCookie:
|     HTTP/1.1 400 Bad Request
|     cache-control: no-cache
|     content-type: text/html; charset=utf-8
|     strict-transport-security: max-age=31536000; includeSubDomains; preload
|     x-content-type-options: nosniff
|     x-frame-options: deny
|     x-xss-protection: 0
|     content-security-policy: default-src 'none'; style-src 'unsafe-inline'
|     <html>
|     <head>
|     <meta content="origin" name="referrer">
|     <title>Bad request &middot; GitHub</title>
|     <meta name="viewport" content="width=device-width">
|     <style type="text/css" media="screen">
|     body {
```

```

| <meta name="viewport" content="width=device-width">
| <style type="text/css" media="screen">
| body {
| background-color: #f6f8fa;
| color: rgba(0, 0, 0, 0.5);
| font-family: -apple-system,BlinkMacSystemFont,Segoe UI,Helvetica,Arial,sans-serif,Apple Colo
| font-size: 14px;
| line-height: 1.5;
| margin: 50px auto; max-width: 600px; text-align: center; padding: 0 24p
|_http-title: Did not follow redirect to https://github.com/
110/tcp open  pop3?
143/tcp open  imap?
443/tcp open  ssl/https github.com
| http-robots.txt: 57 disallowed entries (15 shown)
| /ekansa/Open-Context-Data /ekansa/opencontext-*
| /account-login */tarball/ */zipball/ /Explodingstuff/ /copilot/
| /copilot/c/ /*/*/pulse /*/*/projects /*/*/forks /*/*/issues/new
| /*/*/milestones/new /*/*/issues/search /*/*/commits/
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=github.com
| Subject Alternative Name: DNS:github.com, DNS:www.github.com
| Not valid before: 2026-07-03T00:00:00
|_Not valid after: 2026-09-30T23:59:59
|_http-title: GitHub \xC2\xB7 Change is constant. GitHub keeps you ahead. \xC2\xB7 GitHub
|_http-server-header: github.com
2 services unrecognized despite returning data. If you know the service/version, please submit the

```

TRACEROUTE (using port 80/tcp)

```

HOP RTT ADDRESS
1 0.17 ms 20.207.73.82

```

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 360.92 seconds

Observation:

The aggressive Nmap scan collected more detailed information about the target, including service details, HTTP headers, SSL certificate information, and traceroute results. It also attempted OS and service fingerprinting to gather additional information about the system. Compared to the basic scan, this scan produced much more detailed output, making it useful for a deeper security assessment.

6. Verify whether OpenVAS (Greenbone Vulnerability Manager) is correctly configured in Kali Linux.

```

(student@kali)-[~/Shivam]
└─$ sudo gvm-check-setup
[sudo] password for student:
gvm-check-setup 25.04.0
This script is provided and maintained by Debian and Kali.
Test completeness and readiness of GVM-25.04.0
Step 1: Checking OpenVAS (Scanner) ...
OK: OpenVAS Scanner is present in version 23.38.1.
OK: Notus Scanner is present in version 22.7.2.
OK: Server CA Certificate is present as /var/lib/gvm/CA/servercert.pem.
Checking permissions of /var/lib/openvas/gnupg/*
OK: _gvm owns all files in /var/lib/openvas/gnupg
OK: redis-server is present.
OK: scanner (db_address setting) is configured properly using the redis-server socket: /var
OK: the mqtt_server_uri is defined in /etc/openvas/openvas.conf
OK: _gvm owns all files in /var/lib/openvas/plugins
OK: NVT collection in /var/lib/openvas/plugins contains 95091 NVTs.

```

```

OK: The notus directory /var/lib/notus/products contains 526 NVTs.
Checking that the obsolete redis database has been removed
Could not connect to Redis at /var/run/redis-ovnas/redis-server.sock: No such file or directory
OK: No old Redis DB
Starting ospd-ovnas service
Waiting for ospd-ovnas service
OK: ospd-ovnas service is active.
OK: ospd-OpenVAS is present in version 22.10.0.
Step 2: Checking GVM Manager ...
OK: GVM Manager (gvmd) is present in version 26.15.0.
Step 3: Checking Certificates ...
OK: GVM client certificate is valid and present as /var/lib/gvm/CA/clientcert.pem.
OK: Your GVM certificate infrastructure passed validation.
Step 4: Checking data ...
OK: SCAP data found in /var/lib/gvm/scap-data.
OK: CERT data found in /var/lib/gvm/cert-data.
Step 5: Checking PostgreSQL DB and user ...
Starting postgresql service
OK: PostgreSQL version and default port are OK.
gvmd      | _gvm      | UTF8      | libc      | en_IN     | en_IN     |      |      |
16437|pg-gvm|10|2200|f|22.6||
:   INFO:2026-07-31 06h41.06 utc:28090:   Getting users.
OK: At least one user exists.
Step 6: Checking Greenbone Security Assistant (GSA) ...

```

```

(gsad:28104): GLib-CRITICAL **: 12:11:08.044: g_file_get_contents: assertion 'filename != NULL' failed
OK: Greenbone Security Assistant is present in version Deamon 24.16.0~git.
Step 7: Checking if GVM services are up and running ...
Starting gvmd service
Waiting for gvmd service
OK: gvmd service is active.
Starting gsad service
Waiting for gsad service
OK: gsad service is active.
Step 8: Checking few other requirements ...
OK: nmap is present.
OK: ssh-keygen found, LSC credential generation for GNU/Linux targets is likely to work.
OK: nsis found, LSC credential package generation for Microsoft Windows targets is likely to work.
OK: xsltproc found.
WARNING: Your password policy is empty.
SUGGEST: Edit the /etc/gvm/pwpolicy.conf file to set a password policy.
Step 9: Checking greenbone-security-assistant ...
OK: greenbone-security-assistant is installed

It seems like your GVM-25.04.0 installation is OK.

```

Observation:

The gvm-check-setup command verified that the required OpenVAS components, certificates, databases, and services were installed correctly. Most of the checks returned **OK**, and the final message confirmed that the Greenbone Vulnerability Manager installation was properly configured and ready for use.

7. Start the OpenVAS service and access the Greenbone Security Assistant (GSA) web interface.

```

(student@kali)-[~/Shivam]
$ dpkg -l | grep gvm
ii  gvm                25.04.3      all      remote network sec
ii  gvm-tools          25.4.6-1     all      Remote control the
ii  gvmd               26.15.0-1    amd64    Manager Module of
ii  gvmd-common        26.15.0-1    all      architecture indep
ii  libgvm22t64:amd64  22.35.4-1    amd64    remote network sec
ii  postgresql-18-pg-gvm  22.6.12-1    amd64    PostgreSQL extensi
ii  python3-gvm        26.9.1-1     all      Greenbone Vulnerab

```

```
(student@kali)-[~/Shivam]
$ sudo systemctl status ospd-openvas
● ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd-openvas)
   Loaded: loaded (/usr/lib/systemd/system/ospd-openvas.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-07-31 12:10:33 IST; 13min ago
  Invocation: 51ece00298904b20829f064ef0405e20
     Docs: man:ospd-openvas(8)
           man:openvas(8)
  Process: 27702 ExecStart=/usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/
 Main PID: 27710 (ospd-openvas)
    Tasks: 5 (limit: 4456)
  Memory: 50.1M (peak: 425.1M, swap: 75.2M, swap peak: 75.2M)
     CPU: 43.105s
   CGroup: /system.slice/ospd-openvas.service
           └─27710 /usr/bin/python3 /usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-c
           └─27720 /usr/bin/python3 /usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-c

Jul 31 12:10:33 kali systemd[1]: Starting ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (
Jul 31 12:10:33 kali systemd[1]: Started ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (o
```

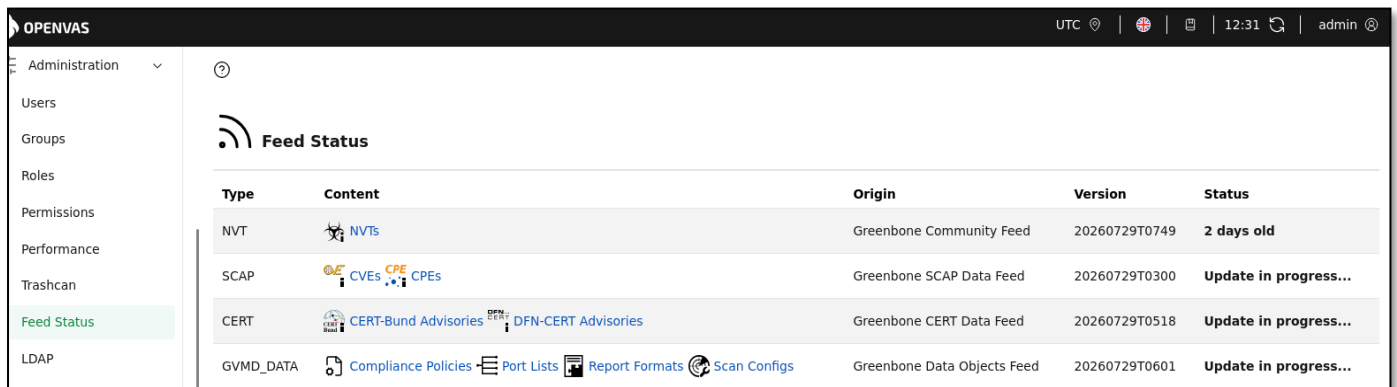
```
(student@kali)-[~/Shivam]
$ sudo systemctl status gvmd
● gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)
   Loaded: loaded (/usr/lib/systemd/system/gvmd.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-07-31 12:11:14 IST; 14min ago
  Invocation: b30daa926f4343bda830e68065b82c61
     Docs: man:gvmd(8)
  Process: 28112 ExecStart=/usr/sbin/gvmd --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-g
 Main PID: 28114 (gvmd)
    Tasks: 3 (limit: 4456)
  Memory: 930.2M (peak: 1.2G, swap: 58.9M, swap peak: 58.9M)
     CPU: 2min 2.403s
   CGroup: /system.slice/gvmd.service
           └─28114 "gvmd: Waiting " --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=
           └─28185 "gvmd: Synchron" --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=
           └─28190 "gvmd: Syncing " --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=

Jul 31 12:11:08 kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (g
Jul 31 12:11:08 kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) aft
Jul 31 12:11:14 kali systemd[1]: Started gvmd.service - Greenbone Vulnerability Manager daemon (gv
```

```
(student@kali)-[~/Shivam]
$ sudo runuser -u _gvm -- gvmd --get-users
: INFO:2026-07-31 06h58.16 utc:65744: Getting users.
admin
```

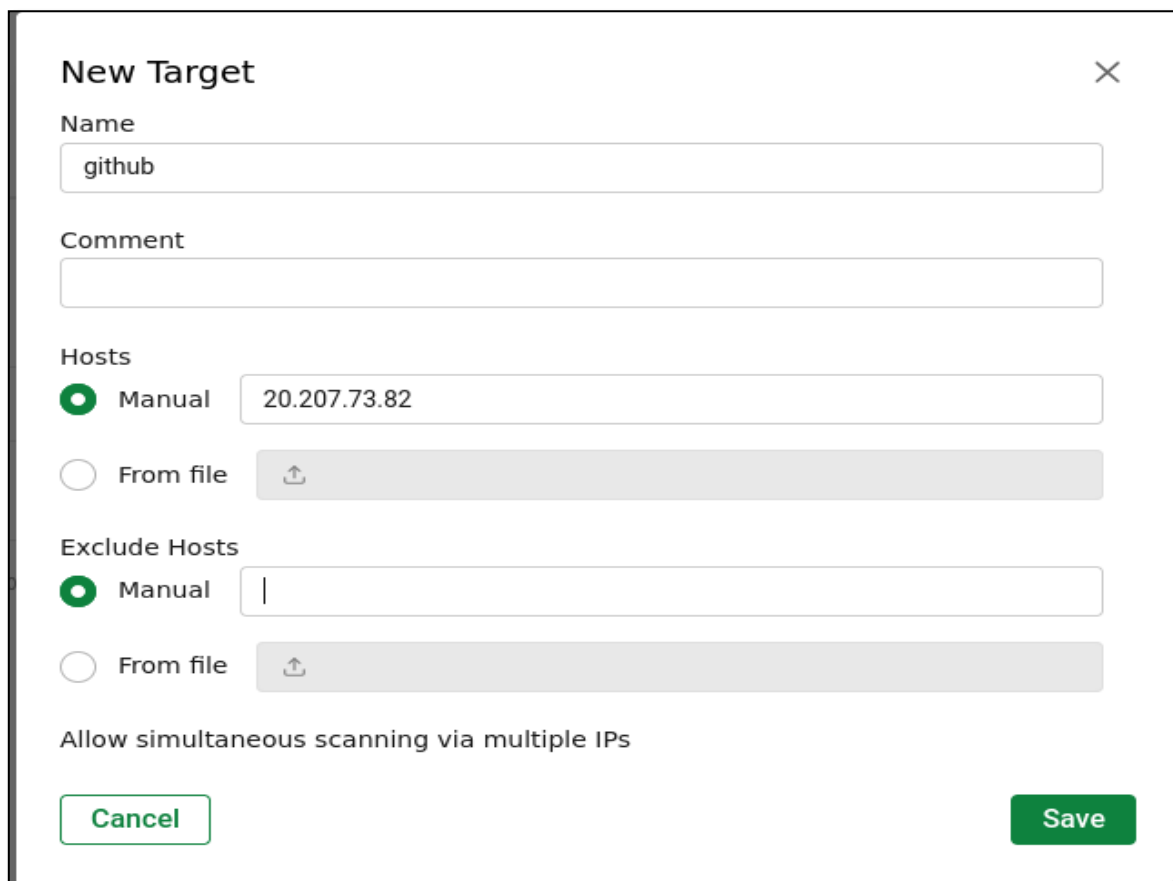
```
(student@kali)-[~/Shivam]
$ sudo runuser -u _gvm -- gvmd --user=admin --new-password=Admin@123
: INFO:2026-07-31 07h01.38 utc:67567: Modifying user password.
```

```
(student@kali)-[~/Shivam]
$ sudo gvm-start
[i] GVM services are already running
```



The OpenVAS Feed Status page displays a table of feeds:

Type	Content	Origin	Version	Status
NVT	NVTs	Greenbone Community Feed	20260729T0749	2 days old
SCAP	CVEs , CPE , CPES	Greenbone SCAP Data Feed	20260729T0300	Update in progress...
CERT	CERT-Bund Advisories , DFN-CERT Advisories	Greenbone CERT Data Feed	20260729T0518	Update in progress...
GVMD_DATA	Compliance Policies , Port Lists , Report Formats , Scan Configs	Greenbone Data Objects Feed	20260729T0601	Update in progress...



New Target

Name:

Comment:

Hosts:

☒ Manual:

☐ From file:

Exclude Hosts:

☒ Manual:

☐ From file:

Allow simultaneous scanning via multiple IPs

Scans

Assets

Resilience

Security Information

Configuration

Targets

Targets 2 of 2

New Task

Auto Delete Reports

☒ Do not automatically delete reports

☐ Automatically delete oldest reports but always keep newest reports

Scanner

OpenVAS Default

Scan Config

Full and fast

Maximum concurrently executed NVTs per host

4

Maximum concurrently scanned hosts

20

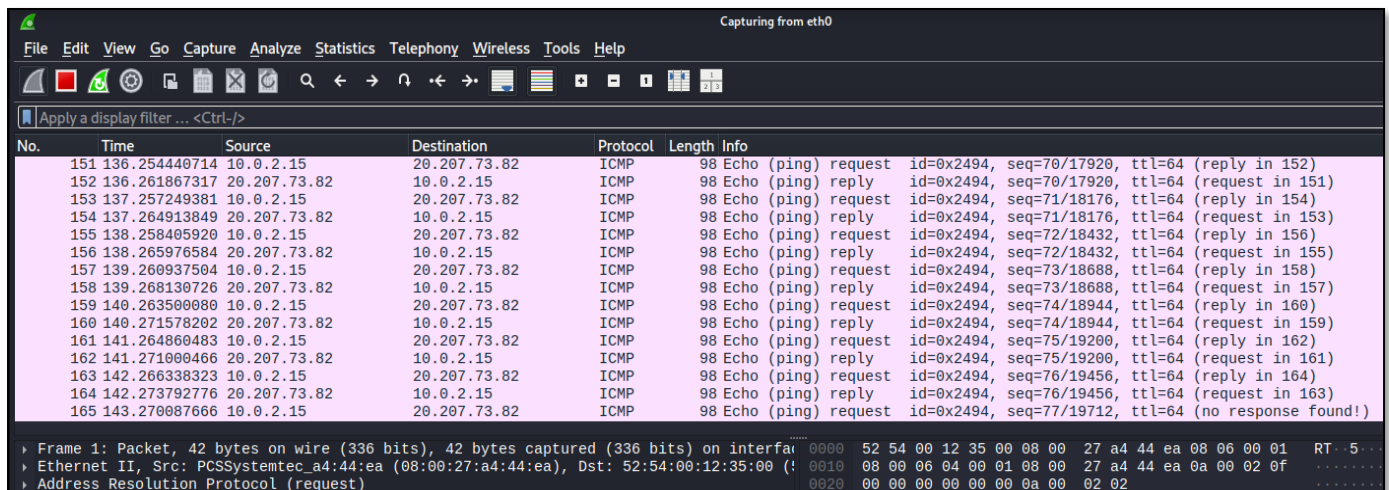
Cancel Save

Observation:

- The output lists the installed **GVM/OpenVAS** packages, confirming that all the required components are available in the Kali Linux system.
- The **osspd-openvas** service is shown as **active (running)**, indicating that the OpenVAS scanner is running successfully and is ready to perform vulnerability scans.
- The **gvmd** service is active and running, confirming that the Greenbone Vulnerability Manager daemon is functioning properly.
- The command displays the available administrator user (admin), confirming that a valid **GVM** user account exists.
- The administrator password was updated successfully, allowing secure login to the **Greenbone Security Assistant (GSA)**.
- The output shows that all GVM services are already running, indicating that the **OpenVAS** environment is fully operational.
- The **Greenbone Security Assistant dashboard** opened successfully, confirming that the web interface is accessible and ready for vulnerability assessment.

- The **Feed Status** page displays the available security feeds and their update status, showing that the **vulnerability database** is synchronized and ready for scanning.
- A new target named **GitHub** was created by entering its IP address, preparing the target system for vulnerability assessment.
- The newly created target appears in the targets list, confirming that it has been saved successfully and is available for future scan tasks.

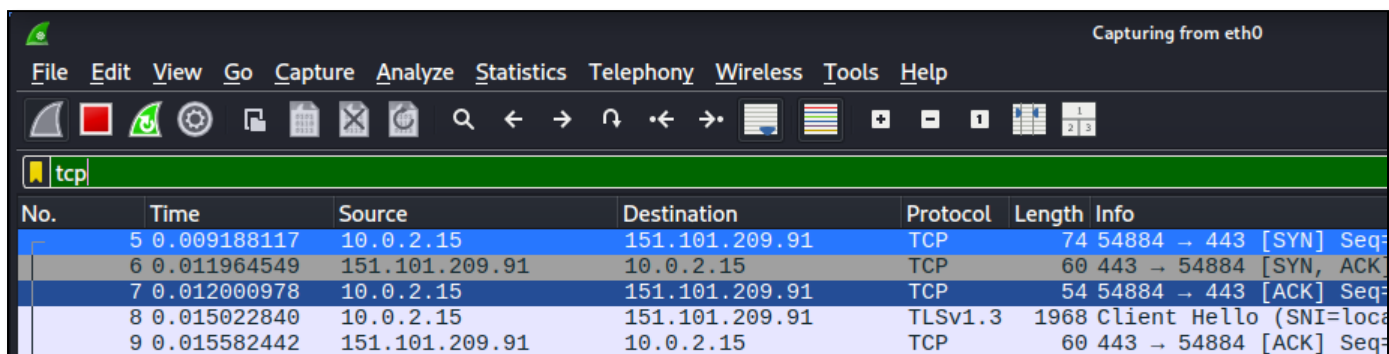
8. Launch Wireshark and select the appropriate network interface for packet capture.



Observation:

Wireshark was launched successfully, and the active network interface was selected for packet capture. After generating network traffic, packets were captured in real time, confirming that Wireshark was monitoring communication on the selected interface correctly.

9. Apply a TCP filter in Wireshark and analyze the captured TCP packets.



Observation:

After applying the TCP filter, only TCP packets were displayed in the capture window. The captured traffic included connection establishment packets such as **TLSv1.3** showing the normal TCP three-way handshake used to establish reliable communication between the client and the server.

10. Apply a DNS filter in Wireshark and identify DNS query and response packets.

No.	Time	Source	Destination	Protocol	Length	Info
3	67.113056653	10.0.2.15	192.168.7.254	DNS	70	Standard query 0x8f0b A github.com
4	67.113147801	10.0.2.15	192.168.7.254	DNS	70	Standard query 0x2e0d AAAA github.c
5	67.121892646	192.168.7.254	10.0.2.15	DNS	86	Standard query response 0x8f0b A gi
6	67.121892906	192.168.7.254	10.0.2.15	DNS	98	Standard query response 0x2e0d AAAA
9	67.128708528	10.0.2.15	192.168.7.254	DNS	85	Standard query 0x82e2 PTR 82.73.207
10	67.135102596	192.168.7.254	10.0.2.15	DNS	85	Standard query response 0x82e2 No s

Observation:

Applying the **DNS** filter displayed DNS query and response packets exchanged between the client and the DNS server. The captured traffic showed successful hostname resolution, where the requested domain name was translated into its corresponding IP address before communication with the target system.

11. Apply an HTTP filter in Wireshark and analyze the HTTP request and response packets.

No.	Time	Source	Destination	Protocol	Length	Info
649	3.166311291	10.0.2.15	151.101.209.91	HTTP	364	GET /success.txt?ipv4 HTTP/1.1
661	3.172863808	10.0.2.15	151.101.209.91	HTTP	364	GET /success.txt?ipv4 HTTP/1.1
664	3.285843546	151.101.209.91	10.0.2.15	HTTP	463	HTTP/1.1 200 OK (text/plain)
3930	14.067609240	10.0.2.15	142.251.42.227	OCSP	481	Request
3932	14.143380440	142.251.42.227	10.0.2.15	OCSP	1240	Response
4007	14.254560589	10.0.2.15	142.251.42.227	OCSP	481	Request
4071	14.336272439	142.251.42.227	10.0.2.15	OCSP	1240	Response

Observation:

The **HTTP** filter displayed only HTTP packets in the captured traffic. The capture included HTTP requests sent by the client and the corresponding responses from the server, including a **200 OK** response, indicating successful communication between the client and the web server.